

Accelerated CGH Generation for AR Head-Up Display Based on DeepCGH and Yolo

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Abstract

This work presents a prototype augmented reality head-up display (AR-HUD) framework that integrates YOLO-based vehicle detection, monocular depth estimation, and phase-only hologram generation using the DeepCGH model. Driving-scene video is processed frame by frame, where detected vehicle regions are converted into binary target patterns for hologram synthesis, and a monocular depth map is employed to estimate the relative distance of each bounding box, enabling depth-aware holographic rendering. The predicted depth information is incorporated into the propagation model to assign appropriate reconstruction distances for different targets. The DeepCGH network generates corresponding phase holograms in real time, forming a holographic sequence that mirrors both the motion and depth variation of vehicles in the original video. Optical validation using an LCoS-based Fourier reconstruction setup confirms temporal consistency and depth correspondence in the reconstructed bounding boxes. Performance evaluation shows stable per-frame operation for both YOLO detection and DeepCGH inference, although reconstruction quality degrades in complex or densely populated scenes. The results demonstrate the feasibility of combining object detection, monocular depth estimation, and computational holography within a unified AR-HUD pipeline, while highlighting future improvements in latency reduction, depth accuracy, and reconstruction fidelity for practical automotive applications.

Keywords

hologram, real time, DeepCGH, YOLO, MiDaS, AR-HUD.

1. Introduction

Augmented reality head-up displays (AR-HUDs) are increasingly explored in advanced driver-assistance systems (ADAS) to present scene-relevant information directly within the driver's forward field of view [1]. To support depth-consistent visualization with minimal distraction, computer-generated holography (CGH) provides wavefront-based 3D reconstruction suitable for windshield-based AR.

However, conventional iterative CGH methods such as Gerchberg-Saxton (GS) require repeated forward-backward propagation for phase retrieval, leading to iteration-dependent latency that is incompatible with frame-by-frame dynamic driving scenes [2]. Learning-based CGH addresses this limitation by replacing iterative optimization with direct neural inference; DeepCGH embeds differentiable propagation into training and enables fixed-complexity, fast phase-only hologram prediction [3–4].

For AR-HUD applications, a key challenge is maintaining consistency across multiple virtual depths. Most existing holographic pipelines optimize or reconstruct at a single plane, whereas depth-aware AR rendering requires multi-plane propagation and structured depth encoding [5–7]. In parallel, recent vision pipelines combine object detection with monocular depth estimation (e.g., YOLO with MiDaS) to infer object-wise relative depth from a single RGB camera in near-real-time settings [8–9]. Yet these perception systems typically stop at geometric

interpretation and are not directly connected to wave-optical hologram synthesis.

Building on these developments, this work integrates YOLO-based vehicle detection and MiDaS depth estimation with a multi-plane DeepCGH framework. Detected vehicle regions are mapped to discrete propagation planes and encoded into a volumetric target representation, forming a unified perception-to-wavefront pipeline for depth-aware AR-HUD holographic visualization.

2. Theory

2.1 Integrated DeepCGH Algorithm

Figure 1 illustrates the overall pipeline. The process begins with video input, where each frame is analyzed using YOLO for vehicle detection and MiDaS for monocular depth estimation. Detected vehicle regions are fused with their estimated depth values and discretized into five predefined depth intervals. Based on this mapping, a five-plane volumetric intensity tensor of size (1024, 1024, 5) is constructed and normalized as the input to the hologram prediction network.

The 5-plane tensor is then fed into a modified DeepCGH model. The U-Net-based architecture with Space-to-Depth enhancement learns a nonlinear mapping from multi-depth intensity targets to a phase-only hologram. During training, angular spectrum method (ASM) propagation is applied to all five depth planes using fixed physical parameters ($\lambda = 532$ nm, pixel pitch = 9.2 μ m, predefined propagation distances). A multi-plane reconstruction loss is computed by comparing propagated intensities with ground-truth depth slices, and gradients are backpropagated through both the optical propagation model and the neural network.

After convergence, the trained model performs single-pass inference without iterative optimization. The input tensor (1024, 1024, 5) is directly mapped to a phase-only hologram (1, 1024, 1024) in one forward pass.

Finally, the predicted phase is numerically propagated to the five depth planes using ASM to generate multi-depth reconstructions. The outputs include the phase hologram and five reconstructed intensity distributions (recon_p0–recon_p4), completing the end-to-end perception-to-hologram AR-HUD pipeline.

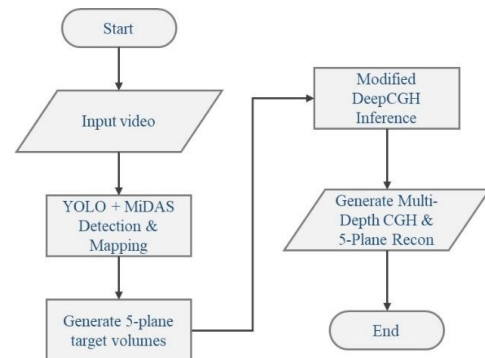


Figure 1. The flowchart of Integrated DeepCGH Algorithm.

2.2 Multi-depth DeepCGH

1. Target Construction

As shown in Fig. 2, a 2D object pattern is generated and assigned to a predefined depth plane according to the multi-depth configuration. The pattern is embedded into a 3D tensor of size (H, W, P), where each slice corresponds to one reconstruction depth. The tensor is normalized to ensure numerical stability during training and optical propagation.

2. Phase Prediction Network

The multi-depth tensor is fed into a U-Net–based network to learn the mapping from volumetric intensity targets to a phase-only hologram. The predicted phase encodes the wavefront modulation required to reconstruct objects at their assigned depths and is compatible with phase-modulation devices such as SLMs.

3. Multi-Plane Optical Propagation

The predicted phase is combined with a uniform amplitude to form a complex optical field. Fourier diffraction is applied to propagate the field to each depth plane, generating reconstructed intensity distributions. The stacked reconstructions form a multi-plane volume for depth-resolved evaluation.

4. Reconstruction Loss and Optimization

Reconstructed intensities are compared with the original multi-depth targets using a similarity metric. The resulting loss is backpropagated through the differentiable propagation model and network, enabling end-to-end optimization of multi-depth phase encoding.

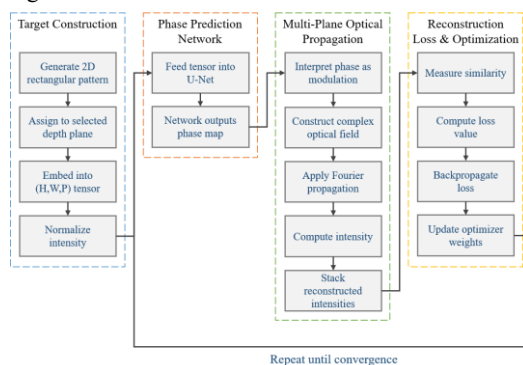


Figure 2. The flowchart of Multi-depth DeepCGH

2.3 YOLOv8+MiDaS

Figure 3 illustrates the perception-to-encoding workflow. First, YOLOv8 performs semantic object detection and outputs bounding boxes $B = [x, y, w, h]$. To improve robustness in driving scenarios, only vehicle-related categories (e.g., cars, trucks, buses, motorcycles) are retained.

Next, a MiDaS model (DPT-Hybrid backbone) generates a dense monocular depth map for the scene. For each detected bounding box, the median depth within the ROI is computed to suppress outliers and improve stability. The depth values are then normalized using percentile scaling to ensure consistent distribution across different scenes.

The normalized depths are discretized into $N = 5$ predefined propagation planes corresponding to the multi-plane holographic configuration. Each object is assigned a discrete depth index, forming a structured volumetric representation.

Finally, the perception results are encoded into (1) a compact feature tensor (1, 8, 7) containing spatial coordinates, size, confidence, and depth index, and (2) a 1024×1024 multi-plane

mask representing spatial occupancy across five depth layers. This dual encoding bridges depth-aware perception with subsequent DeepCGH hologram generation.

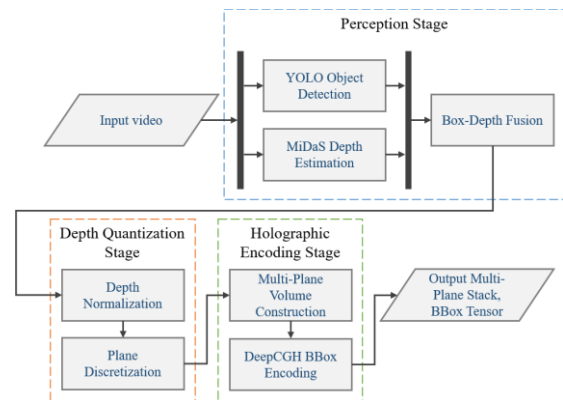


Figure 3. YOLOv8+MiDaS Pipeline.

3. Result & Discussion

Holograms generated by the proposed YOLO–DeepCGH pipeline were validated using a Fourier optical reconstruction setup. A reflective LCoS SLM displayed phase-only holograms under 632.8 nm laser illumination, and reconstruction was performed through a Fourier lens, enabling evaluation of focal clarity and depth consistency under realistic optical conditions.

To verify the multi-depth model, multiple controlled simulation scenes containing spatially separated rectangles at different depths were designed. The corresponding arithmetic depth mappings for different plane spacings are summarized in Table 1, where each equal-interval configuration defines the propagation distances of Planes 0–4. Quantitative evaluation was conducted using the SSIM structure metric across reconstruction planes, with numerical results listed in Table 2. For Plane 4, the 5.0 mm configuration achieves the highest structure value of 0.9746, compared with 0.9714 (2.5 mm), 0.9651 (3.0 mm), and 0.9592 (3.5 mm), demonstrating superior axial selectivity and reduced inter-plane interference.

Among the evaluated scenarios, one representative scene was selected for detailed presentation. The optical projection results further confirm the numerical analysis: when focused on the near rectangle, the far rectangle appears defocused, and vice versa, illustrating clear depth-of-field separation. Compared with smaller spacings (e.g., 3.5 mm), the 5.0 mm configuration exhibits stronger inter-plane separation and more distinct focal transitions.

From a physical optics perspective, the predicted phase encodes propagation-dependent wavefront curvature corresponding to the distances defined in Table 1. Constructive interference occurs at the matched depth plane, while mismatched planes produce defocus blur. The agreement between the SSIM structure values in Table 2 and the optical projection results confirms that the 5.0 mm spacing provides the most balanced focal sharpness and reliable multi-plane depth representation.

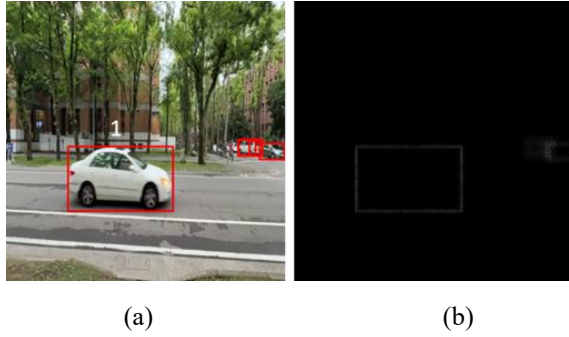


Figure 4. (a) Experimental scenario 3 (b) Simulation results of the rectangle obtained in plane 4.

Table 1. Corresponding arithmetic depth interval mapping.

Plane distance	Plane 0	Plane 1	Plane 2	Plane 3	Plane 4
2.5 mm	5.0	2.5	0.0	-2.5	-5.0
3.0 mm	7.0	4.0	1.0	-2.0	-5.0
3.5 mm	9.0	5.5	2.0	-1.5	-5.0
5 mm	15.0	10.0	5.0	0.0	-5.0

Table 2. Numerical analysis table of Structure values for plane 4.

Plane distance	Rectangle	SSIM	Luminance	Contrast	Structure
2.5mm	rect1	0.1553	0.4348	0.3676	0.9714
3.0mm	rect1	0.1940	0.5061	0.3973	0.9651
3.5mm	rect1	0.1961	0.5106	0.4005	0.9592
5mm	rect1	0.2533	0.5683	0.4573	0.9746

In Fourier-based phase-only holography, conjugate image interference is unavoidable due to the symmetry of the Fourier transform and phase-only modulation of the SLM. To reduce background complexity, This scene was selected for optical validation. Although conjugate overlap remains inherent, evaluation focused on regions minimally affected by superposition. When the system was focused on Rectangle 1 (Fig. 5), the 5.0 mm spacing produced the sharpest reconstruction with clear edge definition, while the distant rectangle appeared defocused. Conversely, when focusing on the far rectangle (Fig. 6), the 5.0 mm configuration yielded the clearest distant reconstruction and visible foreground blur. Compared with smaller spacings (e.g., 3.5 mm), 5.0 mm demonstrated stronger axial selectivity and clearer inter-plane separation.

From a physical optics perspective, the predicted phase encodes depth-dependent wavefront curvature. Constructive interference occurs at the matched propagation plane, while mismatched planes produce defocus blur. Despite conjugate interference, the consistent focal dominance of the 5.0 mm configuration in both near- and far-focus experiments confirms optimal balance between focal sharpness and depth separation, validating the robustness of the proposed multi-depth encoding strategy.

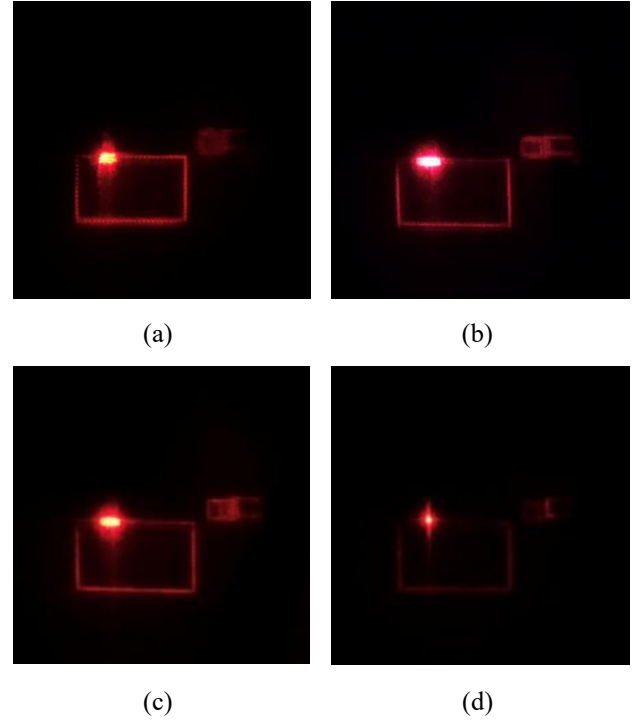


Figure 5. (a) Projected image with a plane distance of 5.0 mm, focused on rectangle 1. (b) Projected image with a plane distance of 3.5 mm, focused on rectangle 1. (c) Projected image with a plane distance of 5.0 mm, focused on rectangle 1. (d) Projected image with a plane distance of 5.0 mm, focused on rectangle 1.

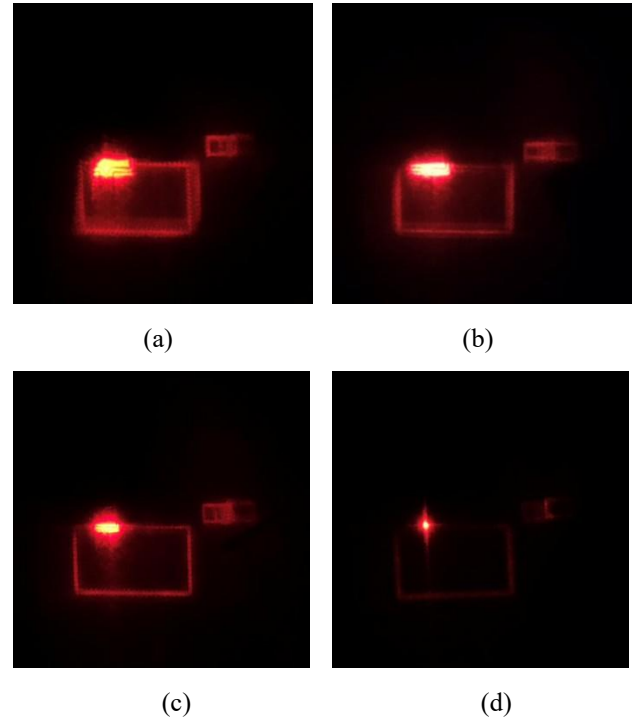


Figure 6. (a) Projected image with a plane distance of 5.0 mm, focused on the far rectangle. (b) Projected image with a plane distance of 3.5 mm, focused on the far rectangle. (c) Projected image with a plane distance of 5.0 mm, focused on the far rectangle. (d) Projected image with a plane distance of 5.0 mm, focused on the far rectangle.

the far rectangle. (d) Projected image with a plane distance of 5.0 mm, focused on the far rectangle.

4. Discussion

The experimental results confirm that the proposed YOLO–MiDaS–DeepCGH pipeline enables depth-aware phase-only hologram generation with consistent temporal behavior. Both simulation and optical validation demonstrate that a 5.0 mm plane spacing provides superior focal sharpness and inter-plane depth discrimination, verifying the correctness of the embedded multi-depth propagation model.

However, performance analysis in Tables 3 and 4 reveals that system-level bottlenecks limit real-time deployment. Although the DeepCGH inference alone achieves 87.12 FPS, full hologram generation including wave propagation drops to 13.99 FPS, indicating that optical propagation dominates computational cost. In the perception stage, YOLO runs at 66.91 FPS, while MiDaS operates at 13.33 FPS, reducing the combined speed to 11.12 FPS. When disk I/O is included, the overall rate further decreases to 3.52 FPS. These results show that depth estimation and data transfer, rather than neural inference, are the primary constraints on throughput.

From an optical standpoint, zero-order diffraction and conjugate image artifacts remain inherent limitations of phase-only Fourier holography, reducing image contrast. Future improvements should focus on artifact suppression and a fully GPU-based tensor-streaming architecture to eliminate disk I/O and redundant memory movement. Lightweight depth models and simplified network design can further increase throughput while preserving the perception-to-wavefront framework.

Table 3. Integrated DeepCGH Algorithm Performance.

Module	Average Time (ms)	FPS
get_hologram	11.479	87.12
CNN-Only Hologram Generation	71.476	13.991

Table 4. Integrated YOLO+MiDaS Performance.

Module	Average Time (ms)	FPS
YOLO	14.954	66.91
MiDaS	75.011	13.33
Total	89.965	11.12
Avg Overall (with saving)	284.485	3.52

5. Conclusions

This study presents an integrated perception-to-hologram multi-depth AR-HUD framework combining YOLO detection, MiDaS depth estimation, and a physics-consistent DeepCGH phase encoding model within a Fourier-based phase-only SLM system. Both simulation and optical experiments validate the embedded multi-plane propagation model. Across all evaluated scenes, SSIM structure analysis consistently identifies 5.0 mm plane spacing as the optimal configuration, achieving balanced focal sharpness and reliable inter-plane depth discrimination.

Optical reconstruction further confirms these findings. Despite

inherent zero-order diffraction and conjugate interference in phase-only Fourier holography, the 5.0 mm configuration exhibits the clearest focal reconstruction and strongest depth-selective behavior. The agreement between numerical SSIM trends and experimental projection verifies correct encoding of propagation-dependent wavefront curvature.

From a system perspective, the DeepCGH network operates efficiently (87.12 FPS), while practical limitations arise from wave propagation, monocular depth estimation, and disk I/O. Future real-time deployment should focus on lightweight depth models and fully GPU-based tensor streaming to reduce latency.

Overall, the proposed YOLO–MiDaS–DeepCGH architecture demonstrates stable depth-aware phase-only hologram generation and provides a scalable foundation for real-time multi-depth AR-HUD visualization.

6. References.

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